

M Sc (BIOTECHNOLOGY/MICROBIOLOGY)
SEMESTER II EXAMINATION (REGULAR)
BT/MI 705 : IMMUNOLOGY

Date: 03.05.2011

Time: 10:00 AM to 1:00 PM

Total Marks: 70

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of non-programmable calculators may be allowed if needed.

PART A

Total marks: 20

Q.1. Choose the correct option and put $\sqrt$ mark in front of it:

1. Adjuvants are **not needed** to enhance immunogenicity when the antigen has the following characteristic

- (a) low immunogenicity (b) small amount (c) high immunogenicity (d) all of above

2. The following epitope is eliminated upon fragmentation by reduction of disulfide bonds in antigen

- (a) non-sequential (b) sequential (c) continuous (d) hydrophilic

3. The following is the characteristic of natural killer cells

- (a) display antigen specificity (b) have T-cell receptors
(c) cytotoxic activity (d) phagocytotic

4. The following organ is **not** supplied by lymphatic vessels

- (a) lymph nodes (b) spleen (c) thymus (d) bone marrow

5. The following molecules are **not** involved in innate immunity

- (a) Lysozyme (b) Interferon (c) Complement (d) none of the above

6. CD8 is a marker of:

- (a) Helper T-cells (b) Cytotoxic T-cells
(c) activated macrophage (d) Neutrophil cells

7. One principal function of the Class I and Class II MHC complex S proteins is to

- (a) transduce the signal to the T-cell interior following antigen binding
(b) mediate immunoglobulin class switching
(c) present antigen for recognition by the T-cell antigen receptor
(d) stimulate production of interleukins

8. The following is true about antigen presenting cells (APC):

- (a) Langerhan's cells are the antigen presenting cells of the epidermis
(b) neutrophils are antigen presenting cells
(c) CD8+ cells only recognize antigen presenting cells bearing MHC class I
(d) TNF α can turn endothelial cells into antigen presenting cells

9. Class II MHC proteins are
 - (a) Recognized by the T4 (or CD4) protein
 - (b) used to mark a cell for killing by cytotoxic T-cells
 - (c) not able to carry an antigen fragment
 - (d) all of above
10. Prior to class switching, B-cells express:
 - (a) IgA
 - (b) IgG and IgA
 - (c) IgD and IgM
 - (d) IgD alone
11. In the complement system;
 - (a) alternative pathway does not rely on antibody
 - (b) classical pathway is best activated by bacterial endotoxin
 - (c) C1 is the first enzyme complex in the classical pathway
 - (d) both the alternative and classical pathway converge at C3
12. The following **is not** involved in evasion of immune response by bacteria:
 - (a) capsule
 - (b) secretion of enzymes
 - (c) Variation in surface Ag
 - (d) none of the above
13. Edema is
 - (a) Accumulation of fluid in injured tissue
 - (b) Release of fluid from injured tissue
 - (c) Oozing of blood from the injured tissue
 - (d) None of the above
14. Lymphoid Immunodeficiencies involves
 - (a) B cells
 - (b) T cells
 - (c) B and T cells
 - (d) None of the above
15. Implanted healthy organ, tissue, or cells are known as
 - (a) Graft
 - (b) Bud
 - (c) Patch
 - (d) Splice
16. Monoclonal Antibodies are produced by single clone of
 - (a) B cells
 - (b) CTL cells
 - (c) Helatocytes
 - (d) Neurons
17. BCG is attenuated in increased concentration of
 - (a) Saliva
 - (b) Bile
 - (c) Serum
 - (d) Cerebrospinal fluid
18. Scatchard plot helps in determining K_a True/ False
19. IDDM is an organ specific autoimmune disease True / False
20. Antigen-antibody reaction is similar to enzyme-substrate reaction that may involve hydrophobic interactions True/ False

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PART B

Total marks: 50

Section I

Q1 (A) Attempt any two of the following: (04)

- (i) Describe Tiselius and Kabat experiment
- (ii) Describe experiments performed for Ab structure elucidation
- (iii) Explain generation of antibody diversity by gene shuffling

Q1. (B) Differentiate between any two of the following: (02)

- (i) Isotypic and idiotypic determinants of immunoglobulin molecules
- (ii) Primary and secondary lymphoid follicles
- (iii) Soluble and membrane bound antibodies

Q1. (C) Describe any two of the following with a schematic diagram: (04)

- (i) Formation of heavy chains of Immunoglobulin by combinatorial joining
- (ii) Mechanism of inflammatory response
- (iii) Mechanism of defense against most pathogens entering through mucosal-associated lymphoid tissue

Q2. (A) Attempt any one of the following: (02)

- (i) During the alternative pathway of complement activation, which event turns C3 tick over into actual activation of complement cascades?
- (ii) What is the role of internal thioester bonds in complement activation?

Q2. (B) Define any two of the following terms: (02)

- (i) ITAM
- (ii) Mature naive B cells
- (iii) Syk Kinase

Q2. (C) Explain briefly any two of the following with a schematic diagram: (04)

- (i) Mode of action of Yop virulence protein in *Yersinia*
- (ii) Mechanisms of surface antigen variation In Influenza virus
- (iii) A tubercle formation in pulmonary tuberculosis

Q2. (D) Justify any one of the following: (02)

- (i) People suffering from neutropenia are susceptible to fungal diseases
- (ii) Waves of parasitemia recur many times in African Sleeping Sickness

Section II

- Q3. (A) Attempt any two** (04)
- In three and four sentences summarize the advantage of having two distinct pathway of antigen processing
 - What are the functions of TAP protein?
 - Why do bigger peptides fit into the peptide binding cleft of the MHC class II molecules then of the MHC class I molecules
- Q3. (B) Write short note on (any one)** (04)
- Humoral immune response
 - Cell mediated immune response
 - Primary and secondary immune response
- Q3. (C) Attempt any one of the following:** (02)
- Draw diagrams illustrating the general structure, including the domains, of class I MHC molecules, class II MHC molecules.
 - Draw diagrams illustrating the general mechanism of class II MHC antigen processing.
- Q4. (A) Write a note on Local Acute Inflammatory Response** (05)
- OR**
- Q4. (A) Write a note on:**
- Systemic Acute-Phase Response (03)
 - SCID (02)
- Q4 (B) Write a short note on any one** (02)
- IDDM
 - Molecular Mimicry
- Q4 (C) Attempt any one:** (03)
- Classify grafts with examples
 - Justify the role of T Cells in allograft rejection
 - Graves' Disease
- Q5. (A) Attempt any one of the following:** (03)
- Why fusant of myeloma cell to myeloma cell dies in the culture during monoclonal antibody production
 - Explain antibody affinity
- Q5. (B) Write short note on any one** (02)
- Antibody avidity
 - DNA Vaccine
- Q5. (C) Write note on any one** (02)
- Live, Attenuated Vaccines
 - Applications of monoclonal antibodies
- Q5. (D) Write note on any one** (03)
- Ouchterlony double immunodiffusion method
 - Toxoid vaccine